数据库系统和信息管理期末复习

规范化

2025 年 6 月考试该题分值为 15 分

必备知识：

1. 阿姆斯特朗定理的使用
2. 各种范式，1NF，2NF，3NF，BCNF，（4NF）定义和其判断方法
3. F^+是什么？K_F^+是什么？如何计算？
4. 候选码如何求？
5. 最小函数依赖集（也就是正则覆盖 canonical cover）？
6. 函数依赖和 ER 图映射基数的关系

大概可能问到以下几个问题：

1. 找出 R 的所有候选码，并解释原因
2. R 是否满足 BCNF/3NF（2NF 总不能让你判断吧）？请给出判断理由。如果不是**范式，请你分解成一组这个范式，并解释你的无损连接和如何在分解的过程中保持函数依赖的
3. 将关系模式分解成某某范式，并指定其主码和外码
3. F 是否是一个规范覆盖（canonical cover）？如果不是，请找出 F 的一个规范覆盖
4. 根据上述约束，写出关系成立的合理函数依赖集 F
5. 找到函数依赖

他妈的注意一下各种缩写是什么意思

1. 阿姆斯特朗定理

自反性：如果 Y⊆X，则 X→Y　一个属性集可以推出它的子集

增广性：如果 X→Y，则 XZ→YZ　给依赖两边加相同属性，依赖仍成立

传递性：如果 X→Y，Y→Z，则 X→Z　类似数学里的传递规则

2. 各种范式

2NF 要满足不存在部分函数依赖（非主属性对码的部分依赖）

3NF 要满足不存在传递函数依赖（非主属性对码有传递依赖 不满足）

BCNF，在非平凡函数依赖的条件下，满足每个决定属性（左边的属性）必须是超码（X→Y，X 必须是超码，在某些题中，超码中一定要有候选码）

3. F^+是什么？K_F^+是什么？

F+是 F 这一组依赖的逻辑蕴含的所有依赖的合集

e.g. F={A→B，B→C}　F+={A→B，B→C，A→C}

K_F^+是 F 下属性集 K 的闭包

e.g. (AB)+={………………}

Fc 是 F 的正则覆盖 canonical cover

4. 求候选码的方法

将依赖关系竖着写成一排，然后分别观察→左右两边的属性

观察要点如下：

左边从来没有出现过什么属性？　称之为 L

右边从来没有出现过什么属性？　称之为 R

哪些属性从来没有出现在左右两边？　　　称之为 N
哪些属性在左右两边都出现了？　　　　　称之为 LR
然后得到以下结论：

R 和 N 一定是候选码中的属性
L 一定不是候选码中的属性
LR 是不确定的，需要我们一个个看

e.g. F={BE→G, BD→G, CD→A, CE→G, CDE→AB, BC→A, B→D}
竖着写成一排：

BE→G
BD→G
CD→A
CE→G
CDE→AB
BC→A
B→D

可以看出，左边没有 A 和 G，右边没有 C 和 E。
所以说，C 和 E 一定存在于候选码中，A 和 G 一定不在候选码中。
我们先从 C, E 看起，(CE)+={C, E, G} 显然是不行的
然后考虑加某个元素形成三个属性构成的候选码

B, C, E　(BCE)+={A, B, C, D, E, G}=F　　B, C, E 是候选码
C, D, E　(CDE)+={A, B, C, D, E, G}=F　　C, D, E 是候选码

可见，候选码有俩

5. 如何求 canonical cover / 正则覆盖 / 最小函数依赖？

分为三步走
第一步：→ 右边单一化
（比如说 A→BC 要化成 A→B 和 A→C）
第二步：从头看看有多有没有多余的项
（比如说有三个依赖是 A→B 和 B→C 和 A→C，显然我们可以看到 A→C 是多余的项）
第三步：看→左边的属性组的闭包是否包含→右边的属性
（比如说 AB→C，看(AB)+={……}中有没有 C 这个元素，有的话说明这个依赖是多余的）
这个意思是如果没有 AB→C 这个条件，还能通过 AB 的闭包退出 C 的话，说明这个依赖是多余的

e.g. F={BE→G, BD→G, CD→A, CE→G, CDE→AB, BC→A, B→D}
第一步：BE→G, BD→G, CD→A, CE→G, CDE→A, CDE→B, BC→A, B→D
第二步：B→D，BD→G，说明 B→D，B→G；
CD→A，CDE→A，说明 CD→A；
B→G，BE→G，说明 B→G；
得到{ B→D, B→G, CD→A, CE→G, CDE→B, BC→A }
第三步：→左边的属性组的闭包是否包含→右边的属性？
如果没有 CD→A，(CD)+={C, D}　不包含 A，说明 CD→A 不多余
如果没有 CE→G，(CE)+={C, E}　不包含 G，说明 CE→G 不多余
如果没有 CDE→B，(CDE)+={A, C, D, E, G}不包含 B，说明 CDE→B 不多余
如果没有 BC→A，(BC)+={A, B, C, D, G}包含 A，说明 BC→A 是多余的
最后 Fc={B→D, B→G, CD→A, CE→G, CDE→B}

6. 判断范式答题模板

2NF：某某关系模式存在非主属性对码的部分依赖，不满足 2NF

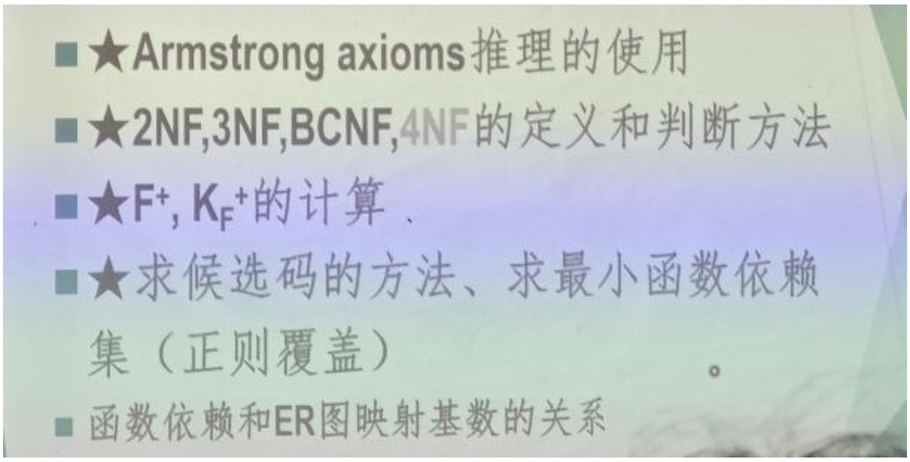
3NF：某某关系模式存在非主属性对码的传递依赖，不满足 3NF

BCNF：某某关系模式的某某依赖是非平凡函数依赖，其左端不含超码，不满足 BCNF

7. 无损连接和保持函数依赖

无损连接：R1∩R2→R1 或 R1∩R2→R2 必须至少有一个成立

函数依赖：F′=F1∪F2∪…∪Fn，F′=F 成立



1. 2022-2023 年考试题

1. Given a relation schema R(A,B,C,D,E) and a set of functional dependencies F={A→B,A→C,B→C,B→E,C→D}, answer the following questions:

(1) Find the candidate keys of R. If you think R has no candidate keys, write "no" (3points)

(2) Is R in BCNF? Please give the reason for your judgment. (3 points)

(3) Is F a canonical cover? if no, please find a canonical cover for F. (4 points)

2. Consider a relation schema StudentProject(sno, sname, deptname, pno, pname, pfund, reward) with the following business rules:

Each student has a unique sno that determines their sname and deptname

Each project has a unique pno that determines its pname and pfund

Each student-project participation (sno, pno) pair has a unique reward

Answer the following questions:

(1) According to the above constraints, please write out the reasonable functional dependencies holding on relation StudentProject（3 points）

(2) List all the candidate keys of StudentProject （2 points）

(3) is StudentProject in 3NF? If it is not in 3NF, please decompse it into a set of 3NF relations and explain why your decomposition is lossless and dependency perseving（5 points）

1．设有关关系模式 R(A,B,C,D,E) 和函数依赖集 F={A→B,A→C,B→C,B→E,C→D}。请回答以下问题:

(1) 找出 R 的所有候选键。如果你认为 R 没有候选键，写“无”。(3 分）候选码是 A

(2) R 是否满足 BCNF? 请给出判断理由。(3 分)

不满足 如 B→C，非平凡，左部不含超码，违反 BCNF 条件。

(3) F 是否是规范覆盖（canonical cover）？如果不是，请找出 F 的一个规范覆盖。(4 分)

不是。 Fc={ A→B, B→CE, C→D }

2. 每个学生有唯一的学号 sno，该学号决定其姓名 sname 和所在系 deptname。

每个项目有唯一的项目号 pno，该项目号决定其项目名称 pname 和项目经费 pfund。

每个学生参与项目的情况由(sno, pno)组合唯一决定，并对应一个酬金 reward。

请回答以下问题:

(1) 根据上述约束，写出关系学生项目上成立的合理函数依赖集 F。(3 分)

F = {sno→sname, sno→deptname, pno→pname, pno→pfund, (sno, pno)→reward}

(2) 列出 学生项目 的所有候选键。(2 分)

候选码：(sno,pno)

(3) 学生项目 是否满足 3NF? 如果不满足，请将其分解为一组满足 3NF 的关系，并解释你的分解为什么是无损连接且保持函数依赖的。(5 分)

不满足 3NF，关系模式存在非主属性对码的部分依赖，不满足 2NF，也就不满足 3NF

用合成法进行分解:

Fc={ sno→sname, deptname, pno→pname,pfund, (sno, pno)→reward }

R1: student(sno,sname,deptname), F1={ sno→sname,deptname }

R2: project(pno,pname,pfund), F2={ pno→pname,pfund }

R3: participate(sno, pno,reward), F3={(sno, pno)→reward }

因为 R1 ∩ R3→R1， R2 ∩ R3→R2 所以分解无损

因为 F1∪F2∪F3=F，所以分解依赖保持 这里注意答题规范，R 是……，F 是……

2. 练习题

1. Consider the following relational schema:

project_material=(prjno, prjname, prjbudget, begindate, enddate, materialNo, materialName,materialquantity)

It contains information about the material usage of projects.

The Attribute prjno is the No. of project, The Attribute prjname is the name of project ,The Attribute prjbudget is the budget of project.

The Attribute materialquantity is quantities of materials used in a projerct.

- **Each project has unique project No.**
- **Different projects may have same project name, project budget, project begindate and project enddate.**
- **Each project use various materials.**

The following is an instance of the schema:

prjNo	prjName	prjBudget	begindate	enddate	Material No	Material Name	Material quantity
101	1# project	10000	2020/01/01	2021/01/01	10001	Cement	100
101	1# project	10000	2020/01/01	2021/01/01	10002	Steel	120
101	1# project	10000	2020/01/01	2021/01/01	10003	Sand	120
102	2# project	20000	2020/09/01	2021/07/01	10001	Cement	100
102	2# project	20000	2020/09/01	2021/07/01	10003	Sand	110
103	1# project	10000	2020/01/01	2021/01/01	10001	Cement	100

(1) **Based on above, Identify functional dependencies of *project_material*.**

(2) Based on above, Identify the **candidate key(s)** of **project_material**.

(3) Is the relation schema **project_material**in **BCNF**? Why? Is it in **3NF**? Why? If it is not in 3NF, bring it to a set of relations at least in 3NF; specify primary keys and referential integrity constraints for each relation.

2.Consider a relation R(A,B,C,D,E,G) with the set of Functional Dependencies

F={BE→G, BD→G, CD→A, CE→G, CDE→AB, BC→A, B→D}

(1) Give all candidate keys of R. [3 points]

(2) Give a canonical cover of F.

(1) Based on above, Identify functional dependencies of project_material. 找到函数依赖

prjNo→prjName, prjNo→prjBudget, prjNo→begindate, prjNo→enddate

materialNo→materialName

prjNo,materialNo→materialquantity 这个可以总结一下（项目和材料，商店和物品，一对多的关系……）

(2) Based on above, Identify the candidate key(s) of project_material.

候选码（prjNo，materialNo）

(3) Is the relation schema project_materialin BCNF? Why? Is it in 3NF? Why? If it is not in 3NF, bring it to a set of relations at least in 3NF; specify primary keys and referential integrity constraints for each relation.

既不是 BCNF，也不是 3NF，因为主码（prjNo，materialNo）对于 prjName 等属性是部分函数依赖，不是 2NF，也不是 3NF 和 BCNF

给出如下 3NF 分解：

R1：project(prjno, prjname, prjbudget, begindate, enddate)， PK:prjno
F1：{prjNo→prjName, prjNo→prjBudget, prjNo→begindate, prjNo→enddate }
R2：material(materialNo,materialName)   PK:materialNo
F2：{materialNo → materialName}
R3： useage(prjno，materialNo，materialquantity) PK:(prjno，materialNo) FK: prjno，materialNo
F3：{(prjNo, materialNo) → materialQuantity}

2.Consider a relation R(A,B,C,D,E,G) with the set of Functional Dependencies
F={BE→G, BD→G, CD→A, CE→G, CDE→AB, BC→A, B→D}
分析：形式化考察规范化理论，函数依赖的闭包，属性的闭包，候选码的计算，最小化函数依赖计算，分解算法等。

(1) Give all candidate keys of R. [3 points]

经过观察单个属性，两个属性的闭包都不可能得到 R，继续看左部 3 个属性的继续查，根据已有左部可以得到这些组合 BED, BEC,BCD,CDE

BED+={BEDG}, BEC+={BECGDA}=R， BCD+={BCDA}，CDE+={CDEABG}=R

没有必要继续找左部 4 个属性的，因为候选码是最小超码。

所以候选码只有 2 个：BCE, CDE

(2) Give a canonical cover of F.

分析：根据求 Fc 的方法：相同左部合并，删除右部多余属性，输出左部多余属性。参见书 上 8.4.3 算法。

考察 F={BE→G, BD→G, CD→A, CE→G, CDE→AB, BC→A, B→D}

注意：这里，因为 BC+由其它函数依赖可以求出包含 A， 所以 BC→A 的右端 A 是冗余的！需要去掉 BC→A 这个函数依赖。 根据教科书 7 版 P231 页第 2 行尾：“如果一个函数依赖在它的右侧只包含一个属性，例如 A→C，并且这个属性被证明是多余的，那么我们将得到一个右侧为空的函数依赖。这样的 函数依赖应该被删除”！

Fc={B→D, B→G, CD→A, CE→G, CDE→B}

(3) Is R in 3NF? explain why if it is or decompose it into 3NF if not.

分析：Not in 3NF, for “CE→G” violate 3NF definition.左边不是码，右边是非码属性（不含候选码属性。

decompose it into

R1({B,D,G}, { B→DG }),

R2({A,C,D}, { CD→A }),

R3({C,E,G}, { CE→G }),

R4({B,C,D,E}, { CDE→B }

也就是根据第二问的正则覆盖搞的

3. 陈鹏练习题

Conside a relational schema R<U, F>:

U={B,O,I,S,Q,D}, F={ S→D, I→B, IS→Q, B→O}

1. Give all candidate keys of R.
2. Give the normal form level of R (until BCNF) and explain why.
3. If R not meets 3NF losslessly decompose it into a set of relational schemas that satisfied 3NF and preserved dependency.

4. 2021-2022 年期末考试题

请为对下列每个关系模式完成:

a. 求出所有的候选码

b. 判定它的范式级别并说明理由.

c. 如果它的范式级别未达到 3NF，请给出一个保持函数依赖的无损分解使其满足第三范式.

1. 运动员 (ID, 姓名, 生日, 团队, 领队电话)

一个运动员只能加入一个团队；一个团队只有一个领队，但一个人可以担任多个团队的领队. 领队只有一个电话.

2. Shop (NO, name, addr, owner, item)

一个商店(shop)有一个店主(owner),一个地址(addr),多个商品(item)

3. R（U，F），U={A, B, C, D}，F={BC, (A,B)D, CB}

打个样:

R<（ID, 姓名，生日，团队，领队电话），{ID→姓名, ID→生日, ID→团队，团队→领队电话}>

a. 候选码是 ID;

b. 不存在非主属性的部分函数依赖，范式级别满足 2NF，但是存在 ID→团队，团队→领队电话的传递依赖，不满足 3NF。

c. 进行如下分解:

R1：队员（ID, 姓名，生日，团队） F1：{ID→姓名, ID→生日, ID→团队}

R2：团队（团队，领队电话） F2：{团队→领队电话}

R1∩R2→R2，是无损分解（这里一定要注意能推导出什么，是 R1 还是 R2）

F1∪F2=F，保持了函数依赖

R<(NO, name, addr, owner, item) {NO→name, NO→addr, NO→ownwe}>

a. 候选码是（NO, item）

b. 因为候选码是（NO, item），(NO, item)→name，存在非主属性的部分函数依赖，不满足 2NF，范式级别为 1NF。

c. 进行如下分解

R1：(NO, name, addr, owner) F1：{ NO→name, NO→addr, NO→ownwe }

R2：(NO, item) F2：Ø

R1∩R2→R1 满足无损分解

F1∪F2=F 保持了函数依赖

5. 2018-2019 年期末考试题

1. Consider the following relational schema:

student_graduation_project=(sID, name, iID, iName, iTitle,year, projectName,grade)

It contains information about graduation designs of students. The Attribute iID is the ID of Instructor, The Attribute iIName is the name of Instructor, The Attribute iTitle is the title of Instructor, We have the following information on the attributes:

- Each student has only one instructor as graduation design advisor.
- Each student has only one graduation design project.
- Each instructor can advisor many students in graduation design.
- Students can take part in same projects.

The following is an instance of the schema:

sID	Name	iID	iName	iTitle	projectName	grade
11	Cheng	101	Yuan	Professor	Face Image Recognition	B
12	Zhang	101	Yuan	Professor	Machine Learning	A
13	Liu	102	Cheng	Instructor	Face Image Recognition	A
14	Yang	102	Cheng	Instructor	Machine Learning	C
15	Li	103	Zhou	Professor	Speech Recognition	D

(1) Based on above, indicate for each of the following potential functional dependencies, whether it is indeed an FD or not. **(3 points)**

A. sID →Name　B. sID→iID　C.iID->grade

D. iName->iID　E. ID→(projectName,grade)　F. projectName→grade

True：A，B，E 选对一个给 1 分

(2) Based on above, indicate for each of following sets of attributes whether it is a candidate key for **student_ graduation_project** or not. **(3 points)**

A. {sID}　B. {sID, iID}　C. {sID,projectName}

D. {name}　E. {name,iName}　F. {iID,projectName}

Candidate Key：A

(3) Is the relation schema **student_ graduation_project** in **BCNF**? Why? Is it in **3NF**? Why? If it is not in 3NF, bring it to a set of relations at least in 3NF; specify primary keys and referential integrity constraints for each relation. **(4 points)**

Not in BCNF and not in 3NF. {sID} is the candidate key of it, iID →iName(其他的也行), iID is not a super key and title is not in any candidate key.

This gives the following 3NF relations:（命名可以不同）

instructor(<u>iID</u>, iName, iTitle)

student_project(<u>sID</u>, name, iID, projectName,grade)　(iID)-> instructor

2. Consider a relation R(A, B, C, D, E) with the set of Functional Dependencies

$F=\{A \rightarrow BD, BC \rightarrow E, B \rightarrow D, D \rightarrow A\}$

(1)Give all candidate keys of R. [3 points]

AC，BC，DC

(2)Give a canonical cover of F.　[3 points] (ANSWER IN NOT UNIQUE)

$Fc=\{A \rightarrow B, BC \rightarrow E, B \rightarrow D, D \rightarrow A\}$

(3)Is R in 3NF? explain why if it is or decompose it into 3NF if not. [4 points]

(ANSWER IN NOT UNIQUE)

No,decompose it into { R1(AB), R2(BCE),R3(BD),R4(DA) }

6. 2017-2018 A

1. Consider the following relational schema:

Articles=(ID, title, journal, issue, year, start_page, end_page, TR_ID)

It contains information on articles published in scientific journals. Each article has a unique ID, a title, and information on where to find it (name of journal, what issue, and on which page). Also, if results of an article previously appeared in a "technical report" (TR), the ID of the technical report can be specified. We have the following information on the attributes:

- For each journal, an issue with a given number is published in a single year.
- The end page of an article is determined by the start page.
- There is never more than one article on a single page.

The following is an instance of the schema:

ID	title	journal	issue	year	start_page	end_page	TR_ID
2	Cuckoo Hashing	JAlg	51	2015	68	80	87
17	Time Serial DM	JAlg	48	2014	69	85	62
17	Time Serial DM	JAlg	48	2014	69	85	78
23	P vs. NP resolved	SICOMP	23	2016	3	7	98
25	P vs. NP resolved	JACM	46	2016	1	5	98

(1) Based on above, indicate for each of the following potential functional dependencies, whether it is indeed an FD or not. **(3 points)**

A. ID →title B. start_page→end_page C. (journal, issue) →year

D. title→ID E. ID→(start_page, end_page, journal, issue) F. TR_ID→ID

True：A，C，E 选对一个给 1 分

(2) Based on above, indicate for each of following sets of attributes whether it is a candidate key for ***Articles*** or not. **(3 points)**

A. {ID} B. {ID, TR_ID} C. {ID, title, TR_ID}

D. {title} E. {title, year} F. {start_page, journal, issue}

Candidate Key：B

(3) Is the relation schema ***Article*** in **BCNF**? Why? Is it in **3NF**? Why? If it is not in 3NF, bring it to a set of relations at least in 3NF; specify primary keys and referential integrity constraints for each relation. **(5 points)**

Not in BCNF and not in 3NF. {ID, TR_ID} is the candidate key of it, ID →title(其他的也行), ID is not a super key and title is not in any candidate key.

This gives the following 3NF relations:（命名可以不同）

Journals(<u>journal, issue</u>, year)

Articles(<u>ID</u>, title, journal, issue, start_page, end_page) FK：(journal, issue)->Journals

Attivle_TRs(<u>ID, TR_ID</u>) FK:ID ->Articles

2. Consider the relation schema R = (A, B, C, D, E, F) and the set of functional dependencies F= {A→B, A→C, BC→E, BC→D, E→F, BC→F}

(1) List the candidate key(s) for R. Write 'none' if you think there are no candidate keys. **(3 points)**

Candidate key: (A)

(2) List the FDs in F that violates BCNF. **(3 points)**

BC→E, BC→D, E→F, BC→F

(3) Is R in 3NF? Why? **(3 points)**

BC→E, BC→D, E→F, BC→F(写一个就行) : LHS is not a super key, and RHS is not in any candidate key.

6. 2017-2018 B

1. Consider the following relational schema:

student_dept=(ID, name, tot_credit, dept_name, building, budget)

It contains information about students and their departments in a university. A student studies in just one department, and a department has exactly a building and budget. The following is an instance of the schema:

ID	name	tot_credit	dept_name	building	budget
00128	Zhang	102	Comp. Sci.	Taylor	100000
12345	Shankar	32	Comp. Sci.	Taylor	100000
19991	Brandt	80	History	Painter	50000
23121	Chavez	110	Finance	Painter	120000
44553	Peltier	56	Physics	Watson	70000
45678	Levy	46	Physics	Watson	70000

(1) Identify **functional dependencies** of ***student_dept*** based on above. **(3 points)**

(2) Identify the **candidate key(s)** of ***student_dept***. **(3 points)**

(3) IS the relation schema ***student_dept*** in **BCNF**? Why? Is it in **3NF**? Why? If it is not in 3NF, bring it to a set of relation schemas at least in 3NF; specify primary keys and referential integrity constraints for each relation. (5 points)

2. Consider the relation schema R = (A, B, C, D, E) and the set of functional dependencies F= {A→B, C→D, AC→E}

(1) List the candidate key(s) for R. Write 'none' if you think there are no candidate keys. **(3 points)**

(2) List the FDs in F that violates BCNF. **(3 points)**

(3) Is R in 3NF? Why? **(3 points)**

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1. Given a relation schema R(A,B,C,D,E) and a set of functional dependencies $F=\{A\rightarrow B, A\rightarrow C, B\rightarrow C, B\rightarrow E, C\rightarrow D\}$, answer the following questions:

(1) Find the candidate keys of R. If you think R has no candidate keys, write "no" (3points)

(2) Is R in BCNF? Please give the reason for your judgment. (3 points)

(3) Is F a canonical cover? if no, please find a canonical cover for F. (4 points)

2. Consider a relation schema StudentProject(sno, sname, deptname, pno, pname, pfund, reward) with the following business rules:

Each student has a unique sno that determines their sname and deptname

Each project has a unique pno that determines its pname and pfund

Each student-project participation (sno, pno) pair has a unique reward

Answer the following questions:

(1) According to the above constraints, please write out the reasonable functional dependencies holding on relation StudentProject（3 points）

(2) List all the candidate keys of StudentProject （2 points）

(3) is StudentProject in 3NF? If it is not in 3NF, please decompse it into a set of 3NF relations and explain why your decomposition is lossless and dependency perseving （5 points）

Conside a relational schema R<U, F>:

$U=\{B,O,I,S,Q,D\}$, $F=\{S\rightarrow D, I\rightarrow B, IS\rightarrow Q, B\rightarrow O\}$

1. Give all candidate keys of R.
2. Give the normal form level of R (until BCNF) and explain why.
3. If R not meets 3NF losslessly decompose it into a set of relational schemas that satisfied 3NF and preserved dependency.